Milestone One: Exploring Socioeconomic Well-Being in Municipal Regions in British Columbia

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Introduction & Question

The British Columbia (BC) Socioeconomic Index (SEI) is a dataset provided by BC Stats which uses measures of income, health, education, and community cohesion to develop a socioeconomic well-being indicator by municipality or community health service area (BC Stats 2026). The economic domain captures income, employment, and housing indicators. The education domain captures educational attainment and course completion. The health domain captures prevalence of health conditions. Lastly, the community domain captures social cohesion through measures such as crime rates, marital status, and rates of income assistance. Each of these four indicators have been consistently used in social health literature as meaningful determinants of health and well-being (please see the methodology report for more information). The SEI is a accumulation of the four domains, with a mean of 50 and standard deviation of 10.

The dataset provides two primary models, a longitudinal model and a detailed model. For the purposes of this project, I will be using the detailed model which contains SEI variables for 2023. Although in a different analysis, the longitudinal model could provide interesting findings on temporal social well-being trends. The SEI data is accessed through proprietary code provided by BC Stats which allows researchers to get various provincial datasets. The SEI allows for meaningful sociological analysis, giving a holistic measure of social well-being across municipal regions in the province. As such, I plan to use this dataset to undertake a exploratory descriptive analysis of SEI scores across municipal regions. To achieve this, I will ask the following research question:

RQ: How does socioeconomic index differ across different municipal regions in British Columbia?

This question will allow me to conduct an exploratory analysis using the Tidyverse to view the SEI in a table and various graph formats (Wickham et al. 2019). There are many patterns

that could be found in the data that would answer the question. One primary finding is how well-being is reported across various municipal regions, which could help policy-makers direct resources or funding to municipalities that score low in one of the domains or have a low SEI in general. Moreover, we could also use this data to look a rural/urban divide trends, examining whether there exist discrepancies in SEI reports among urban or rural communities. Most interestingly to me, I want to see how the SEI differs across two major urban centers in BC - the Greater Victoria Region and the Metro Vancouver Regional District. All of these potential findings can be extrapolated from this data through basic descriptive analysis.

Plan

Please see attached PDF in assignment or GitHub repository.

Data

```
bcdc_search("socioeconomic")
```

The code above is proprietary to 'bcdata' and allows me to search for datasets using key terms such as "socioeconomic" or "public health" (Teucher et al. 2021).

```
#bc_ses_data <- bcdc_get_data("b-c-socioeconomic-index")
```

Uncomment the code above to use it, the document will not render if it not commented. Select "2" to get Census Subdivision Data. This type of data aggregates regions into municipalities, not health regions, which are usually smaller (BC Stats 2026).

Simulate

The first simulation randomly assigns values from 1-100 in domains of income (economics), education level, population health variables, and community cohesion variables. The SEI takes scores on each of these domains to calculate an overall simulated index (BC Stats 2026). The second simulation separates SEI values by either Victoria or Vancouver regional districts and gives us an average SEI score for each regional district. These scores will differ in the actual data.

```r
set.seed(1738)

simulated_sei_data <- tibble(
  Municipality = c("Victoria", "Saanich", "Cental Saanich", "Oak Bay",
           "North Saanich", "Sidney", "Langford", "View Royal",
           "Colwood", "Esquimalt", "Metchosin",
           "Vancouver", "Richmond", "Burnaby", "Surrey", "Delta",
           "New Westminister", "North Vancouver", "West Vancouver",
           "Coquitlam"),
  Economics = sample(1:100, size = 20),
  Education = sample(1:100, size = 20),
  Health = sample(1:100, size = 20),
  Community = sample(1:100, size = 20),
  SEI = rowMeans(data.frame(Economics, Education, Health, Community))
)

simulated_sei_data
```

```r
set.seed(1738)

simulated_regional_sei_data <- simulated_sei_data |>
  mutate(
    Region = recode_values(
      Municipality,
      c("Victoria", "Saanich", "Cental Saanich", "Oak Bay", "North Saanich",
           "Sidney", "Langford", "View Royal", "Colwood", "Esquimalt",
        "Metchosin") ~ "Greater Victoria Region",
      c("Vancouver", "Richmond", "Burnaby", "Surrey", "Delta",
           "New Westminister", "North Vancouver", "West Vancouver",
        "Coquitlam") ~ "Vancouver Metropolitan Area"
    )
  ) |>

group_by(Region) |>
  summarise(
    AverageSEI = mean(SEI)
  )

simulated_regional_sei_data
```

Now we can use Tinytable to print cleaner tables of the simulated data (Arel-Bundock 2026). The first table shows SEI for discrete municipalities and the second table shows the mean SEI

for the Vancouver and Victoria regional districts respectively.

```
simulated_sei_data |>
  tt()
```

Table 1: Simulated SEI Data

Municipality	Economics	Education	Health	Community	SEI
Victoria	53	54	99	29	58.75
Saanich	33	96	66	28	55.75
Cental Saanich	98	29	25	79	57.75
Oak Bay	74	69	10	45	49.50
North Saanich	71	53	100	50	68.50
Sidney	38	4	77	65	46.00
Langford	100	93	24	52	67.25
View Royal	52	5	95	25	44.25
Colwood	51	67	69	94	70.25
Esquimalt	39	91	58	16	51.00
Metchosin	23	7	1	74	26.25
Vancouver	57	21	28	32	34.50
Richmond	15	42	62	84	50.75
Burnaby	19	34	39	53	36.25
Surrey	64	73	19	75	57.75
Delta	6	51	52	10	29.75
New Westminister	93	82	87	77	84.75
North Vancouver	40	75	78	30	55.75
West Vancouver	36	15	80	67	49.50
Coquitlam	1	22	36	60	29.75

```
simulated_regional_sei_data |>
  tt()
```

Table 2: Average Simulated SEI Data by Regional District (Victoria and Vancouver)

Region	AverageSEI
Greater Victoria Region	54.11364
Vancouver Metropolitan Area	47.63889

References

Arel-Bundock, Vincent. 2026. *Tinytable: Simple and Configurable Tables in 'HTML', 'LaTeX', 'Markdown', 'Word', 'PNG', 'PDF', and 'Typst' Formats*. https://doi.org/10.32614/CRAN.package.tinytable.

BC Stats. 2026. *BC Socioeconomic Index Methodology Report and Data*. Province of British Columbia. https://www2.gov.bc.ca/assets/gov/data/statistics/people-population-community/socio-economic/bcstatsnewslettermay2026socioeconomicindex.pdf.

Teucher, Andrew C., Sam J. Albers, and Stephanie L. Hazlitt. 2021. "Bcdata: An r Package for Searching and Retrieving Data from the b.c. Data Catalogue." *Journal of Open Source Software* 6 (61): 2927. https://doi.org/10.21105/joss.02927.

Wickham, Hadley, Mara Averick, Jennifer Bryan, et al. 2019. "Welcome to the tidyverse." *Journal of Open Source Software* 4 (43): 1686. https://doi.org/10.21105/joss.01686.